PRANESH KARTHI M S

Software Engineer

Erode, Tamil Nadu, India | 8270426785 | praneshkarthims@gmail.com | linkedin.com/in/mspk5196 | github.com/mspk5196 | mspk.in | mspkapps.in

SUMMARY

Software engineer and AI & Data Science undergraduate who designs, ships, and maintains production systems end to end. Built a multi-tenant task platform handling 128,000+ daily API calls for 347 users, a school ERP serving 2,315 students with ₹1.74 Cr+ in fees tracked, and a self-published authentication-as-a-service npm package powering SSO across multiple live applications. Strong in Node.js, React/React Native, and SQL databases, with a focus on secure multi-tenant architecture (RBAC/ACL) and automated Docker/Jenkins deployments.

EDUCATION

B.Tech, Artificial Intelligence and Data Science | Expected 2028

Bannari Amman Institute of Technology, Erode — CGPA: 7.84 / 10

TECHNICAL SKILLS

Programming Languages: JavaScript, Java

Frontend Development: React, React Native, Zustand

Backend Development: Node.js, Express.js, RESTful API Design, JWT, OAuth 2.0, WebSockets, Socket.io

Databases & Caching: PostgreSQL, MySQL, Firebase, Redis

System & Security Architecture: Multi-Tenant Architecture, Software Architectural Design, RBAC/ACL, AES-256 Encryption

DevOps & Infrastructure: Docker, Linux, Nginx, Jenkins CI/CD, Cloudflare, Grafana

Tools & Practices: Git, GitHub, Razorpay API, Firebase Cloud Messaging, node-cron, Production Deployment

PROJECTS

MSPK Cloud Storage | *Personal* | *Jun 2026 – Present (In Development)* | cloud.mspkapps.in

- Founded and developed a self-hosted cloud storage platform, in development for 3+ months ahead of launch, keeping files on users' own hardware.
- Created an AES-256-CTR encryption model and specified the desktop Storage Connector architecture using Go + Wails v2.
- Directed development of a React dashboard and companion Android/iOS app, connected to MSPK Auth Service for single sign-on.

School ERP | *School Administration & Finance Platform* | *Mar 2026 – Jun 2026 (In Production, Maintained)*

- Constructed a school administration ERP over a 4-month cycle, serving 2,315 active students across grades 1–12.
- Automated dues calculation and receipts via Razorpay, eliminating manual reconciliation for ₹1.74 Cr+ in tracked fees.
- Enforced role-based access across 4 user tiers with audit logging capturing 1,590+ tracked staff actions.
- Configured the backend API using Express 5 and Sequelize/MySQL with Socket.io updates, containerized via Docker and Nginx.
- Delivered a React 19 web portal and Android/iOS mobile client for administrators, staff, and parents.

TaskApp | *Enterprise Multi-Tenant Task & Workflow Platform* | *Jan 2026 – Aug 2026 (In Production, Maintained)* | 3+ live organizations

- Architected a multi-tenant task platform live for 4+ months across 3 organizations, serving 500+ users.
- Implemented per-organization data isolation and a hierarchical escalation engine for unresponsive tasks.
- Engineered 4 workflow types — direct assignment, sequential packages, recurring schedules, and task-bidding — plus a leaderboard scoring engine.
- Developed the REST API using Node.js/Express/MySQL and a React + Vite admin portal, with CI/CD pipelines via Docker and Jenkins.
- Launched a React Native mobile companion with FCM notifications and Google OAuth SSO for Android and iOS.
- Instrumented an API analytics dashboard monitoring 128,000+ daily calls across 4 organizations and 347 users at 98.8% reliability.

Authentication-as-a-Service Platform | *Personal* | *Nov 2025 – Jan 2026 (Production, Maintained)* | authservices.mspkapps.in

- Published an Authentication-as-a-Service platform as a public npm package, in production since January 2026 across 3 self-built applications.
- Integrated JWT authentication, Google OAuth 2.0 SSO, and Redis session caching for 30–40 active users.
- Deployed the service via Docker, Nginx, Cloudflare Tunnel, and Jenkins CI/CD for continuous, one-command releases.